

Ravi Gyani

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PROFESSIONAL SUMMARY

Hands-on Software Engineer with 20+ years of experience designing and operating cloud-native, event-driven systems at scale. Deep expertise in AWS, serverless architecture, and large-scale data processing. Consistent track record of cutting costs, eliminating technical debt, and shipping high-impact solutions across e-commerce, automotive, media, and national security domains.

TECHNICAL SKILLS

AWS: Serverless & compute (Lambda, ECS, Batch, Step Functions, API Gateway) | Data & analytics (S3, Glue, Athena, OpenSearch, Kinesis) | AI/ML (Bedrock, SageMaker) | IoT & edge (IoT Core, Greengrass) | Security (Inspector, IAM, Cognito)

Infrastructure as Code: Terraform, AWS CDK, CodePipeline — full lifecycle from development to production

Big Data & Streaming: Kafka, Spark, Flink, Airflow, Hadoop, HBase, Redis

Containers & Orchestration: Docker, Kubernetes (KEDA), Prometheus, Grafana

Languages: Python, Java, Golang

CAREER HIGHLIGHTS

Delivered AI-First Search Experience — Tradebyte (Zalando)

- Designed an **NLP-driven interface for OpenSearch**, replacing traditional GUI filters with natural language queries, improving operator efficiency significantly.
- **Fine-tuned a seq2seq model** on Claude 3-generated training data, deployed via auto-scaling ECS for low-latency inference. Automated the full model lifecycle: dataset generation (Bedrock/S3), fine-tuning, containerization, and deployment – enabling rapid iteration as schemas evolve.

Replaced Monolith with Event-Driven Microservices — Tradebyte (Zalando)

- Rearchitected a legacy monolith to a cloud-native, event-driven architecture on AWS Kinesis, DynamoDB, and Kubernetes (KEDA) enabling real-time data processing and fault isolation.
- Cut stock recalculation latency **from 30 minutes to under 2 seconds** precomputing across distributed nodes.
- Consolidated fragmented upstream article-status streams into a single authoritative source of truth, **reducing operational costs by 75%** (\$108k annual saving).
- BEngineered an automated reconciliation processor **using Bloom filters** to recover from upstream event loss, reducing system recovery time from 7 days of manual effort to under 2 hours.
- Protected DynamoDB write costs against a **50x traffic spike** from legacy upstream sources by implementing a **Leaky Bucket** throttling layer, smoothing bursty ingestion into controlled, predictable write throughput.

Eliminated a Database Dependency with Stateless RBAC - Luxoft GmbH

- Replaced a DB-backed permission system with Cognito custom attributes embedded in the JWT — removing an entire API Gateway + database round-trip while maintaining tamper-proof security.

Reduced AMI Provisioning from 1 Day to 5 Minutes - Luxoft GmbH

- Automated Windows EC2 provisioning via Step Functions — cutting a full-day manual process to under 5 minutes, with auto-shutdown on CPU idle for cost savings.
- Implemented SSM Managed Instance Core, eliminating the need for public IPs entirely - turning a provisioning tool into a secure, self-service platform adopted across multiple teams.

Built Federated ML Pipeline for Automotive Fault Detection - Luxoft GmbH

- Implemented Flower federated learning with AWS IoT Core, ECS, and Greengrass to train models across distributed vehicle telemetry without centralizing sensitive data, enabling real-time fleet-wide fault detection.

Slashed Cloud Compute Costs by 90% - Luxoft GmbH

- Inherited a costly batch-processing workflow using Airflow, EFS, S3, and AWS Batch.
- Re-architected using Step Functions, Spot Instances, and S3 runtime mounts, cutting compute costs by 90% while improving reliability and scalability.

Eliminated 99% of Container CVEs via Automated Pipeline - Luxoft GmbH

- Designed an automated ECR image rebuild and promotion pipeline: AWS Inspector → EventBridge → Step Functions → CodePipeline, with automated testing, output comparison, notification, and safe production promotion.

Scaled National Security Analytics to Billions of Location Records - Polaris Wireless

- Replaced a failing SQL Server-based location tracking system with a custom big data platform (Hadoop, HBase, Kafka, Spark), delivering real-time subscriber profiling at national scale.

WORK EXPERIENCE

Principal Engineer — Tradebyte (Zalando), Munich Jan 2025 – Present

- Architecting the migration from a tightly coupled monolith to a distributed, event-driven system, unlocking faster feature delivery and improved fault isolation.
- Leading AWS cloud adoption: spearheading POCs, migration blueprints, and aligning technology choices with business cost and performance objectives.
- Shaping strategic technical direction, balancing long-term scalability with near-term business growth.

Senior Solutions Architect — Luxoft GmbH, Munich Jan 2021 – Jul 2024

- Embedded consultant across Amazon, BMW, Volkswagen, and Continental, owning end-to-end cloud architecture from design through delivery.
- Covered the full spectrum: migration strategy (BMW connected vehicle platform), ML pipeline design (VW), cost optimization, security automation, and IaC (Continental).

Solutions Architect — Polaris Wireless, Bangalore Feb 2014 – Aug 2020

- Architected a scalable location analytics platform for national security, leading the big data transformation from SQL Server to Hadoop, HBase, and Spark.
- Invented and patented a real-time subscriber profiling and location estimation system (US20230050646A1).

Software Engineer & Founder — Various / KNOCKS Solutions 2003 – 2014

- Built iOS/Android, .NET WinForms, and C# applications across multiple client engagements.
- Founded KNOCKS Solutions (2006–2009); shipped a commercial .NET WinForms UI component suite including Outlook-style calendars, VS-style docking, and Office 2007 ribbon controls.

EDUCATION

Bachelor of Engineering - Netaji Subhas Institute of Technology, Delhi | 1999 – 2003

CERTIFICATIONS

- AWS Certified Solutions Architect – Professional (SAP-C01) | Oct 2020
- AWS Certified Security Specialty (SCS-C01) | Mar 2023
- Confluent Certified Developer for Apache Kafka | Jul 2019
- Databricks Certified Developer: Apache Spark 2.X | Jul 2019

PATENTS

Real-Time Subscriber Profiling & Location Estimation System - US20230050646A1

Invented and filed during tenure at Polaris Wireless; enables scalable real-time analytics across billions of location records for national security applications.